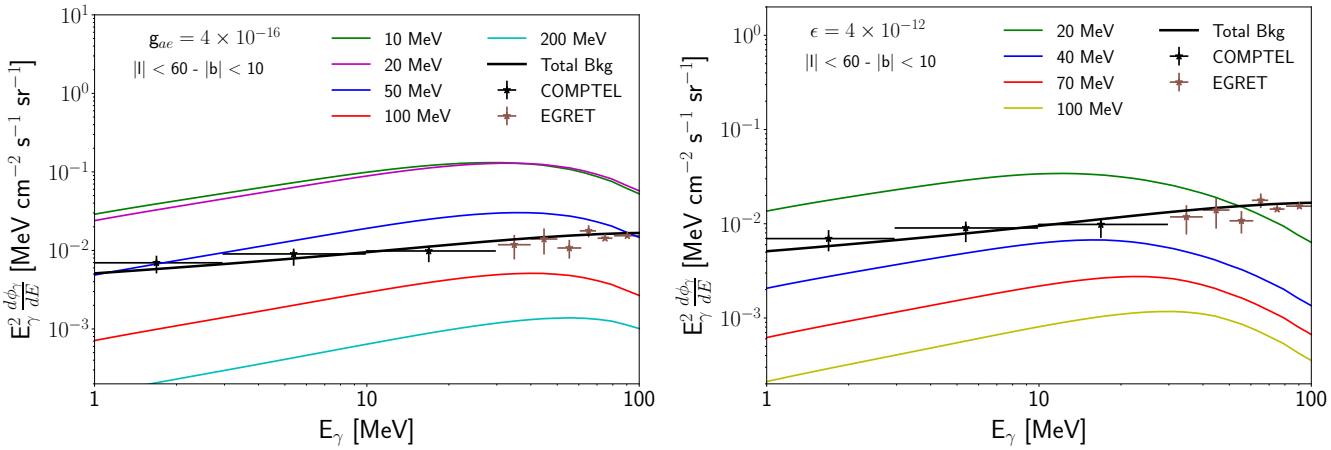

FIG. 5. IA emission from ALPs coupling to baryons and electrons (left panel) and DPs (right panel) produced in SNe, for several masses of these FIPs. For the ALP scenario, production is set by fixing the ALP-proton coupling at $g_{ap} = 2 \times 10^{-11}$. The predicted signals are compared to COMPTEL [108, 109] and EGRET [110] measurements of the diffuse γ-ray flux in the MeV range for a region of interest covering the Galactic plane ($|l| < 60^\circ$ and $|b| < 10^\circ$). The black line in both panels refers to the expected Galactic background emission (from Inverse-Compton and bremsstrahlung of CR electrons).

have recently done in Refs. [24, 111, 112], where we refer the reader for technical details (in particular, see Eq. (2)+ of Ref. [24] and the related discussion). In this calculation, we compute the diffuse 511 keV line emission from the positrons injected by FIPs as proportional to the thermalized distribution of positrons which we assume to follow the steady-state solution for the injected positrons. In addition, to account for the quick decrease in the free-electron density above and below the disk of the Galaxy we apply a scaling relation to the 511 keV profiles, following the vertical distribution of free-electron density in the Galaxy, adopting the NE2001 model [113, 114].

As discussed in Refs. [25, 27], the morphology of the expected 511 keV line emission from the FIP positrons cannot explain the high emission around the center of the Galaxy, leading instead to a flatter profile following the Galactic disk emission, as can be seen from Fig. 5 of Ref. [26].

In the case of the IA emission, we calculate this contribution following our previous work (see Eq. (2) of Ref. [24]), that adopts a similar strategy as in Ref. [115]. This strategy consists of first evaluating the integrated 511 keV line emission in a region of interest and then computing the continuum IA flux via the measured ratio of the para-positronium emission to IA emission at 511 keV. In this way, the energy dependence of the IA emission depends on the probability of relativistic positrons interacting with a free-electron (described by the usual Dirac cross sections [116]) and the time that a positron of a certain energy spends in the Galaxy before thermalizing (i.e. the energy loss rate), while the normalization is fixed by the 511 keV emission in that region of interest. Full details are given in our companion work [24] as well as in a related recent work [112].

Fig. 5 shows the IA emission from ALPs coupling to baryons and electrons (left panel) and DPs (right panel) produced in SNe, for several masses of these FIPs, and compared to COMPTEL [108, 109, 117] and EGRET [110] measurements of the diffuse γ-ray flux in the MeV range. This region of interest provided the best constraints on the sterile neutrino mixing angle in Ref. [24], among the few regions where COMPTEL data of the diffuse Galactic emission is available. This is because this is an extended region that covers the Galactic plane, where SNe can inject these particles. The morphology of the IA signals from ALPs and DPs is similar, although we note that for DPs this emission peaks around $20 - 30$ MeV, while the emission from ALPs peaks at slightly higher energies, around ~ 60 MeV. These features suggest that COMPTEL can be considered a valuable experiment to look for these kinds of signals. Above ~ 60 MeV we start to have a contribution from π^0 decays, making it more difficult to spot any signature from IA emission. Conversely, we note that EGRET is provided with very low spatial resolution, and it is likely to have an important contribution from unresolved sources in their observations of the diffuse Galactic flux. Therefore, we use EGRET data just for illustrative purposes and make use of COMPTEL data in the region $|l| < 60^\circ$ and $|b| < 10^\circ$ to derive constraints. In this figure, we also show the expected background Galactic emission, which mainly consists of IC emission and bremsstrahlung from CR electrons, which are the dominant γ-ray production processes below energies of a few hundred MeV. For this background, we use the electron model from Refs. [118, 119], which is optimized to reproduce the electron and positron emission at Earth location, as well as the local γ-ray emissivity measured by the Fermi-LAT telescope, down to a few tens of MeV.